

Final Examination

(Jan. 10th, 2011)

Name & ID #:

Score:

1. (15pts) A carry-bypass adder is composed of several 4-bit Ripple-Carry Adders (RCA), which is shown in Figure 1. P is the carry propagate signal for each 4-bit cell. Assume one bit adding time of RCA is $T_A=2$, and P signal generation time is $T_P=2$. Delay of AND-gate is $T_1=1$, and delay of OR-gate is $T_2=1$. Derive the worst case propagation delay of this bypass adder and explain in detail.

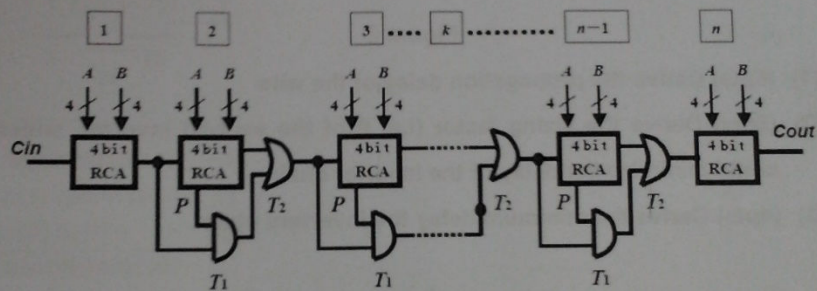
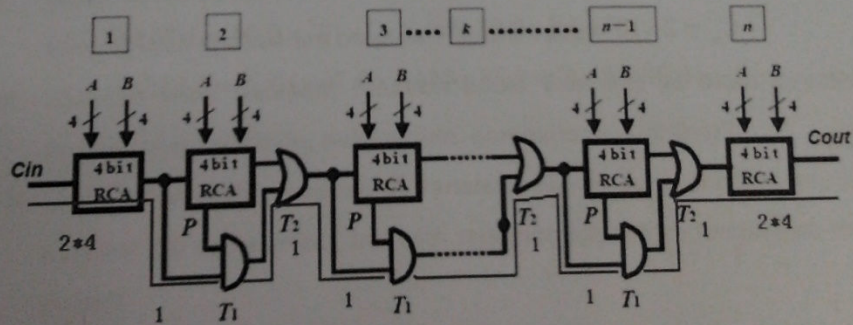


Figure 1

Solutions:

图 8 分，公式 7 分。



The critical time of adder calculation is $8 \times 2 + 2 \times (n-2) = 12 + 2n$. [$n \geq 2$]

2. (15pts) Two inverters are cascaded to drive a metal wire whose length is 2cm and width is $3.6\mu\text{m}$, as illustrated in Figure 2. The first stage inverter is the minimum-sized inverter with $C_g=10\text{fF}$ (gate Cap.), $C_{\text{int}}=10\text{fF}$ (intrinsic Cap.) and $t_{p0} = 175\text{ps}$ (intrinsic delay). The parallel-plate and fringing Caps. of the metal wire are $0.03\text{fF}/\mu\text{m}^2$, $0.04\text{fF}/\mu\text{m}$ respectively, and the sheet resistance of the metal wire is $0.08\Omega/\square$.

注意fringe cap要算两个侧面，乘2

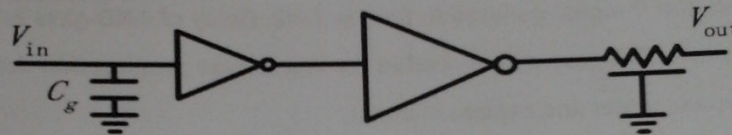


Figure 2

- (1) (6pts) Derive the propagation delay of the wire
- (2) (5pts) Derive the sizing factor (i.e. f) of the second inverter, which could minimize the delay of the inverter chain.
- (3) (4pts) Derive the minimum delay the inverters chain.

这里算的是反相器链的延时

Solution:

1)

$$R_w = 0.08 \times \frac{2 \times 10^{-2}}{3.6 \times 10^{-6}} = 444.4\Omega$$

$$C_w = 2 \times 10^4 \times 3.6 \times 0.03\text{fF} + 2 \times 2 \times 10^4 \times 0.04 = 3.76\text{pF}$$

$$t_w = 0.38 \times R \times C = 0.38 \times 444.4 \times 3.76 \times 10^{-12} = 0.635\text{ns}$$

2)

$$f = \sqrt{\frac{3.76\text{pF}}{10\text{fF}}} = 19.4$$

$$t = 2 \times 175 \times (1 + 19.4) = 7.14\text{ns}$$

3. (15pts) $X=1001011$ and $Y=1100111$ are 2's complement 7-bit binary data, in which the most significant bit is the sign bit. Use modified radix-4 booth encoding approach to calculate the product of X (multiplicand) and Y (multiplier). Please list every partial product of the multiplication and write down the corresponding booth encoding value (such as 0, +X, -X, +2X, -2X) next to it.

Solution:

编码 8 分, 竖式 7 分。

$X=-53$ $Y=-25$

$2X=10010110$ $-2X=01101010$ $-X=0110101$

Extend $Y=111001110$

1 1 1 0 0 1 1 1 0	
0 2X -X	-X
-2X	

00000000110101 -X

111110010110 2X

0001101010 -2X

00010100101101

10100101101=1325

4. (15pts) The circuit of Figure 3 implements a 1-bit datapath function in dynamic (precharge/evaluate) logic.

- (1) (6pts) Write down the Boolean expressions for outputs F and G . On which clock phases are outputs F and G valid?
- (2) (2pts) To what datapath function could this unit be most directly applied (e.g., addition, subtraction, comparison, shifting)?
- (3) (4pts) What is the purpose of transistor M_1 ? Is there another way to achieve the same effect, but with reducing capacitive loading on the clock?

- (4) (3pts) How can the evaluation phase of F be sped up by rearranging transistors? Note that no transistors should be added, deleted, or resized.

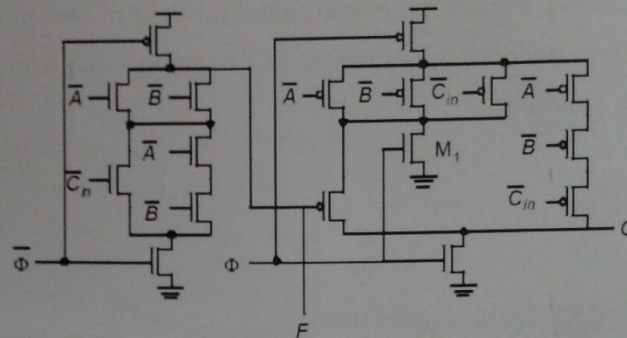


Figure 3

Solution:

- (1) The Boolean function of F is

$$F = \overline{(\overline{A}\overline{B} + \overline{C}_{in})}(\overline{A} + \overline{B})$$

$$= AB + AC_{in} + BC_{in}$$

The Boolean function of G is

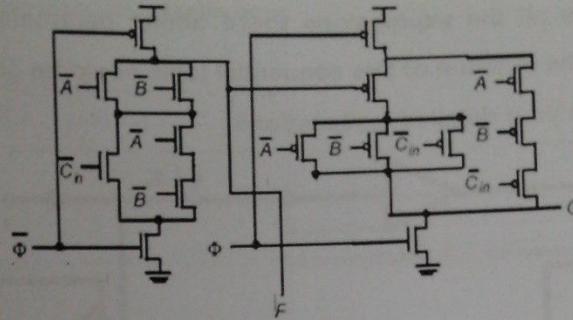
$$G = \overline{(\overline{A} + \overline{B} + \overline{C}_{in})}(\overline{A}\overline{B}\overline{C}_{in} + F)$$

$$= A \oplus B \oplus C_{in}$$

F and G are valid when clock is low.

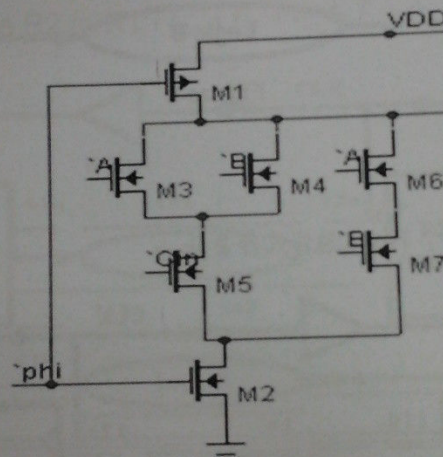
- (2) This bit-slice can be directly used to implement addition function.

- (3) The $M1$'s function is discharging the voltage applied between the internal node capacitors to ground during pre-discharging phase. There is another way to achieve the same purpose, the circuit is shown below:



(4) Speed up the evaluation phase of F by rearranging transistors:

$$F = \overline{(\overline{AB + C_{in}})(\overline{A + B})} = \overline{AB + (\overline{A + B})C_{in}}$$



5. (10pts) Figure 4 shows a sequential logic circuit as shown. Timing parameters of the circuit are as follows:

- Setup time of R1 = t_{1s} , Hold time of R1 = t_{1h} , Propagation delay of R1 = t_{1q}
- Setup time of R2 = t_{2s} , Hold time of R2 = t_{2h} , Propagation delay of R2 = t_{2q}
- Maximum propagation delay of Logic A = t_{a1}
- Minimum propagation delay of Logic A = t_{a2}
- Maximum propagation delay of Logic B = t_{b1}
- Minimum propagation delay of Logic B = t_{b2}
- The clock skew between the clocks of R1 and R2 is δ ($\delta \geq 0$)

Please derive all the expressions these timing parameters should satisfy, so that the function of this sequential logic could be guaranteed. It's better to give your detailed explanations.

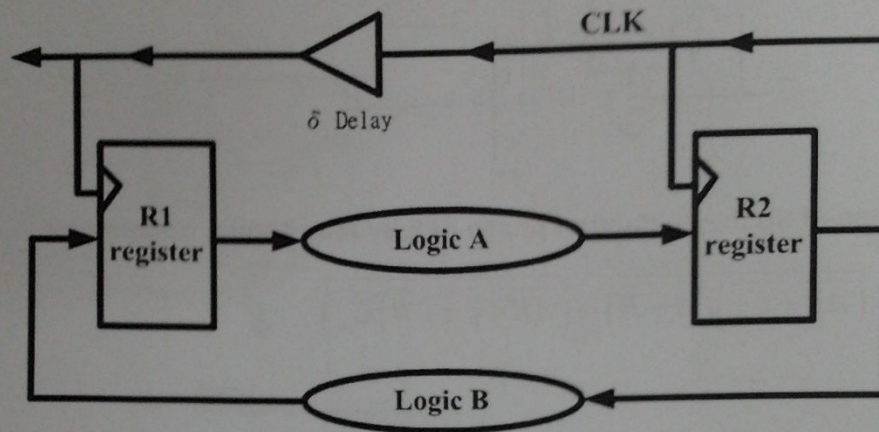
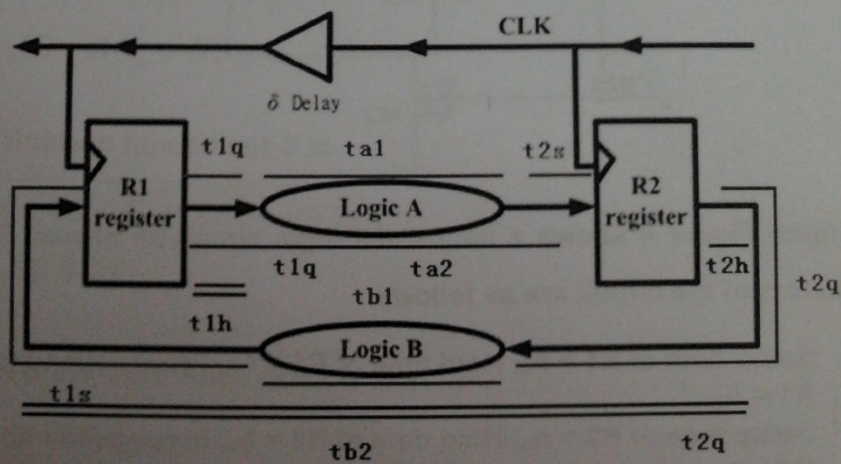


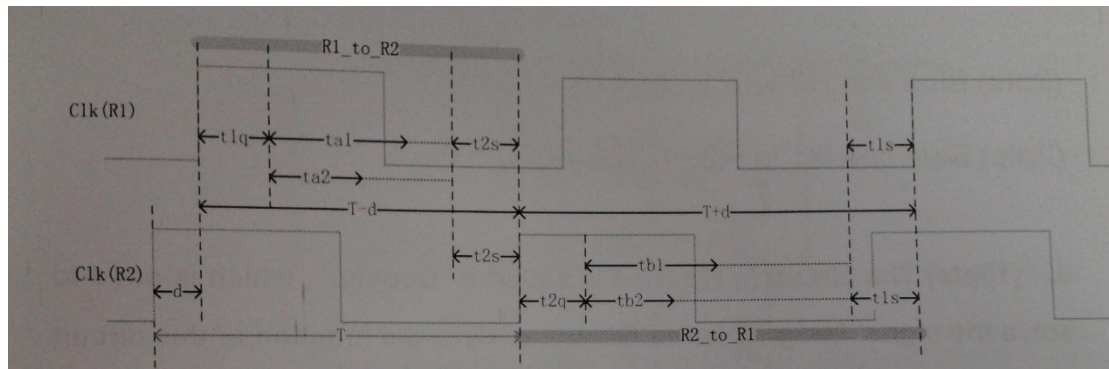
Figure 4

Solution:

Setup 的公式各 3 分, Hold 的公式各 2 分。



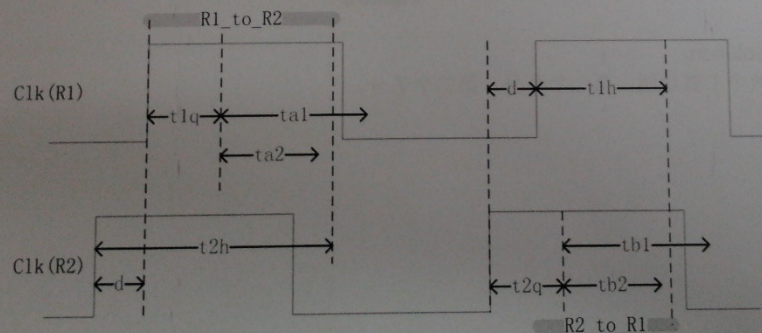
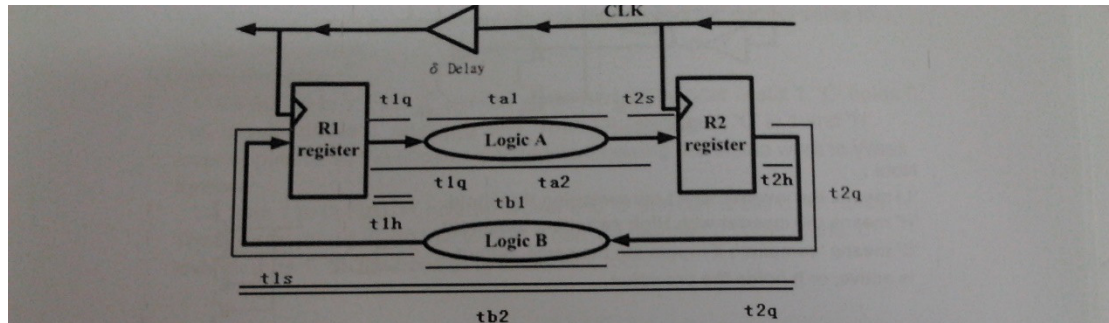
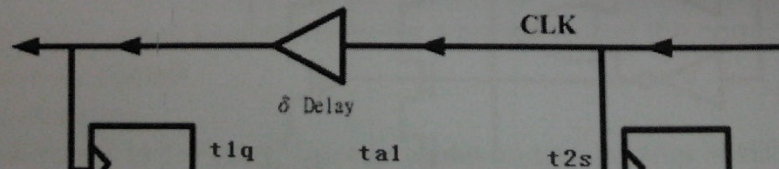
Timing requirements:



For setup time:

(3pts) Red line R1 to R2: $t_{1q} + t_{a1} + t_{2s} \leq T - \delta$

(3pts) Red line R2 to R1: $t_{2q} + t_{b1} + t_{1s} \leq T + \delta$

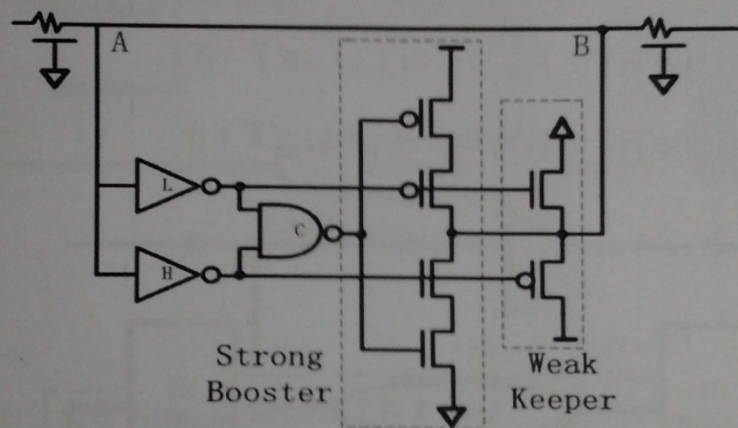


For hold time:

(2pts) Blue line R2 to R1: $t_{2q} + t_{b2} \geq t_{1h} + \delta$

(2pts) Blue line R1 to R2: $t_{1q} + t_{a2} \geq t_{2h} - \delta$

6. (15pts) The circuit in Figure 5 is called a “Booster”, which is used to solve the signal integrity issue. Please analyze the function of this circuit in detail. Note that the length of the wire between node A and node B tends to be very big.



Note :

‘L’ means the inverter with Low switching threshold.

‘H’ means the inverter with High switching threshold.

‘C’ means ‘compare’, if inputs are the same, output is active, or it holds the pre-value.

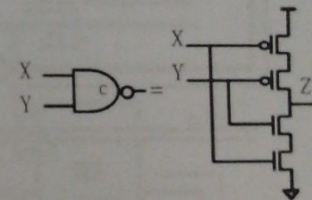
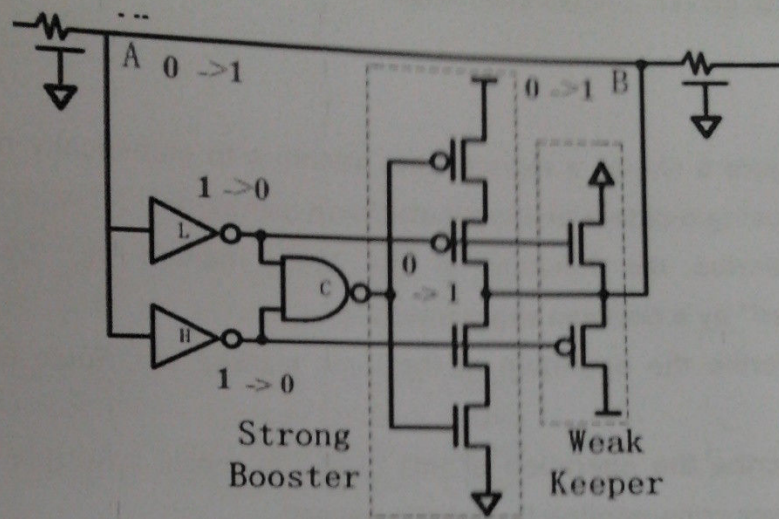


Figure 5

Solution:

两个主要功能，第一个 8 分，第二个 7 分。



When V_A holds, Weak Keeper will keep V_B .

When V_A varies, V_B will be pulled up/down rapidly to the correct values by Strong Booster.

When stable, weak keeper works. It's a buffer, driving the correct value to overcome crosstalk.

When changing,

V_A from 0 to 1: First 'L' inverter works and 'H' inverter is still 1. 'C' holds 0. The booster drives '1' on the 'B' end. Then V_A reaches 1, 'C' is 1 and 'H' inverter puts 0. So the booster is shutdown, leaving the stable work to Weak Keeper.

V_A from 1 to 0: First 'H' inverter works and 'L' inverter is still 0. 'C' holds 1. The booster drives '0' on the 'B' end. Then V_A reaches 0, 'C' is 0 and 'L' inverter puts 1. So the booster is shutdown, too.

(1) (8pts) Describe the function of the logic marked as "Data-Transition Look-Ahead" by a Boolean expression.

(3) (3pts) Describe the operation of this circuit in detail. Why this register can reduce power consumption in statistic sense?

Note: symbol  has the function that when “CKIB” equals 0, the input of this symbol cannot influence the output, otherwise the inverted version of the input will be copied to the output.

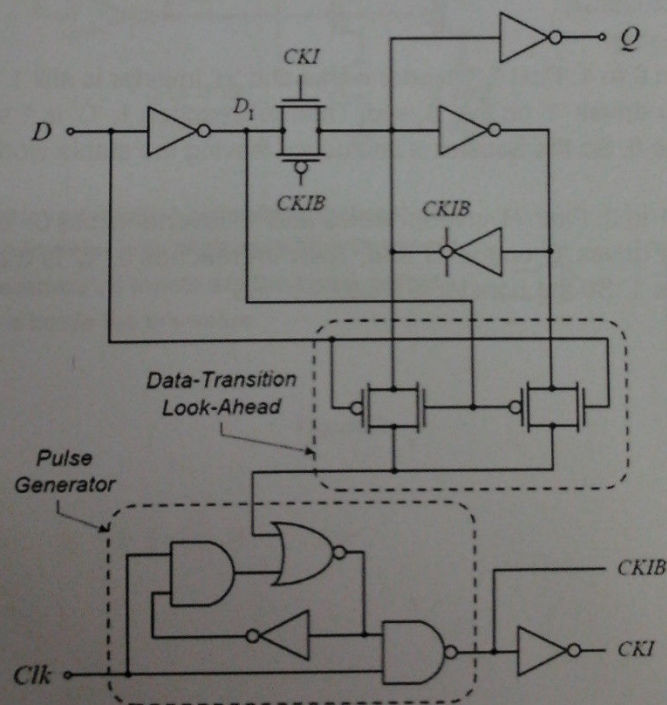


Figure 6

Solution:

(1) signal DL is the result of $(D \text{ XNOR } Q)$. $DL = DQ + D'Q'$

If (CLK==1) and (DL 1 -> 0), then CKIB=1, CKI=0
 If (CLK==1) and (DL 0 -> 1), then CKIB=1, CKI=0
 If (CLK==0) and (DL 1 -> 0), then CKIB=1, CKI=0
 If (CLK==0) and (DL 0 -> 1), then CKIB=1, CKI=0

If (DL==0) and (CLK 0 -> 1), then CKIB=0, CKI=1
 If (DL==0) and (CLK 1 -> 0), then CKIB=1, CKI=0
 If (DL==1) and (CLK 0 -> 1), then CKIB=1, CKI=0
 If (DL==1) and (CLK 1 -> 0), then CKIB=1, CKI=0

In fact, when DL==1, CKIB is always equal to 1 and CKI is always equal to 0 no matter CLK equals 0 or 1. (If DL==1), then CKIB=1, CKI=0).

From the above analysis, we can find out that only when D is not equal to Q, and at that time CLK makes a transition from 0 to 1, then CKIB equals 0 and CKI equals 1.

(3) The CKI/CKIB only transit when D is different from Q with data-transition look-ahead circuitry. The probability of D equal to Q is 0.5, so the switching probability is half reduced.

8. (2pts) Radix-8 Booth Multiplier adopts 3 bits of the multiplier (e.g. in the expression "A x B", "A" is the multiplicand and "B" is the multiplier), together with 1 extra "Observe Bit", to carry out the multiplication. Please derive and prove the encoding law of the radix-8 Booth Multiplier. From your point of view, what are the advantages and disadvantages compared with the Radix-4 Booth Multiplier? (Supplementary Problem. No Penalty for Wrong Solution)?

Solution:

The multiplication operation can be rewritten as following

$$\begin{aligned}
 X \times Y &= X \times (Y_{n-1} \times 2^{n-1} + \dots + Y_0) \\
 &= X \times ((Y_{-1} + Y_0 + 2 \times Y_1 - 4 \times Y_2) + \dots \\
 &\quad + (Y_{n-1} + Y_n + 2 \times Y_{n+1} - 4 \times Y_{n+2}) \times 2^n) \quad n=3K, \\
 &\quad + (Y_{n-2} + Y_{n-1} + 2 \times Y_n - 4 \times Y_{n+1}) \times 2^n) \quad n=3K+1, \\
 &\quad + (Y_{n-3} + Y_{n-2} + 2 \times Y_{n-1} - 4 \times Y_n) \times 2^n) \quad n=3K+2,
 \end{aligned}$$

From the derivation above, the coding rule of radix-8 booth multiplier is:

$$Y_{i-2} + Y_{i-1} + 2 \times Y_i - 4 \times Y_{i+1}$$

advantages:

Partial product is generated for every 3 bits, the product number is further reduced, speeds up the multiplication.

disadvantages:

3X is required during the computation which cannot be implemented by shifting, and this will make the circuit more complex which may offset the advantage brought by lower partial product number.